

# PBAE-PEG-based lipid nanoparticles for lung cell-specific gene delivery

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**Exemplified by successful use in COVID-19 vaccination, delivery of modified mRNA encapsulated in lipid nanoparticles (LNPs) provides a framework for treating various genetic and acquired disorders. However, LNPs that can deliver mRNA into specific lung cell types have not yet been established. Here, we sought to determine whether poly( $\beta$ -amino ester)s (PBAE) or PEGylated PBAE (PBAE-PEG) in combination with 4A3-SC8/DOPE/cholesterol/DOTAP LNPs could deliver mRNA into different types of lung cells *in vivo*. PBAE-PEG/LNP was similar to Lipofectamine MessengerMAX followed by PBAE/LNP for mRNA transfection efficiency in HEK293T cells *in vitro*. PBAE-PEG/LNP administered by intravenous (IV) injection achieved 73% mRNA transfection efficiency into lung endothelial cells, while PBAE-PEG/LNP administered by intratracheal (IT) injection achieved 55% efficiency in lung alveolar type II (ATII) epithelial cells in mice *in vivo*. PBAE/LNP administered by IT injection were superior for specific delivery into lung airway club epithelial cells compared to PBAE-PEG/LNP. Lipofectamine MessengerMAX was inactive *in vivo*. 5-Methoxyuridine-modified mRNA was more efficient than unmodified mRNA *in vivo* but not *in vitro*. Our findings indicate that PBAE-PEG/LNP and PBAE/LNP can transfect multiple lung cell types *in vivo*, which can be applied in gene therapy targeting genetic lung diseases.**

## INTRODUCTION

Genetic and acquired disorders affecting lung function contribute to the global burden of lung disease. Mutations in genes critical for normal functions of pulmonary epithelial, endothelial, and mesenchymal cells cause a number of debilitating conditions, including cystic fibrosis, surfactant protein disorders, ciliary dyskinesia, and pulmonary phospholipid clearance disorders. An intact form of a number of genes, including SFTPC,<sup>1</sup> SFTPB,<sup>2</sup> NKX2-1,<sup>3</sup> ABCA3,<sup>4</sup> CFTR,<sup>5</sup> TBX4,<sup>6</sup> and FOXF1,<sup>7</sup> is essential for lung formation or function.<sup>8</sup> Mutations or dysregulation of these genes cause life-threatening diffuse pulmonary disorders in infants and children.

Recent advances in nanoparticle development have shown significant promise. Technologies such as organ-targeting lipids (e.g., selective organ targeting [SORT]),<sup>9–12</sup> preformed vesicles (PFV)-lipid nanoparticles (LNPs),<sup>13</sup> and polymer-LNPs<sup>14–16</sup> present substantial potential for gene therapy. SORT LNPs, for instance, exhibit remarkable lung-targeting ability, enhanced by increased 1,2-dioleoyl-3-trimethylammonium propane (DOTAP) to improve lung targeting via intravenous (IV) administration and include ionizable amino lipids and helper lipids to enhance transfection efficiency. SORT LNPs have demonstrated both lung endothelial and epithelial transfection ability.<sup>9–11</sup> Similarly, SM-102 was shown to improve lung transfection efficiency.<sup>12</sup> Poly( $\beta$ -amino ester)s (PBAEs) have also been tested for delivering DNA and mRNA into lungs by intratracheal (IT) or IV injection.<sup>17,18</sup> The combination of PBAE and LNPs comprising 1,2-dioleoyl-*sn*-glycero-3-phosphoethanolamine (DOPE)/cholesterol/C14-polyethylene glycol (PEG) was tested for delivering mRNA into lung endothelial cells by IV injection.<sup>15</sup> These LNPs and PBAE exhibit low toxicity, biocompatibility, and biodegradability compared with polyethylenimine-based and poly-L-lysine-based nanoparticles.<sup>12,19,20</sup> However, the combination of PBAE-PEG (or PBAE) and SORT lipids (4A3-SC8/DOPE/cholesterol/DOTAP; hereafter, LNP) to create PBAE-PEG/LNP or PBAE/LNP has not been extensively explored. In the present study, we synthesized two distinct types of PBAEs, PBAE-PEG and PBAE, and combined them with LNPs to produce two unique nanoparticle formulations that we tested *in vitro* and *in vivo*. IV injection of both PBAE-PEG/LNP and PBAE/LNP selectively delivered mRNA into pulmonary endothelial cells. IT administration of these nanoparticles demonstrated that PBAE-PEG/LNP

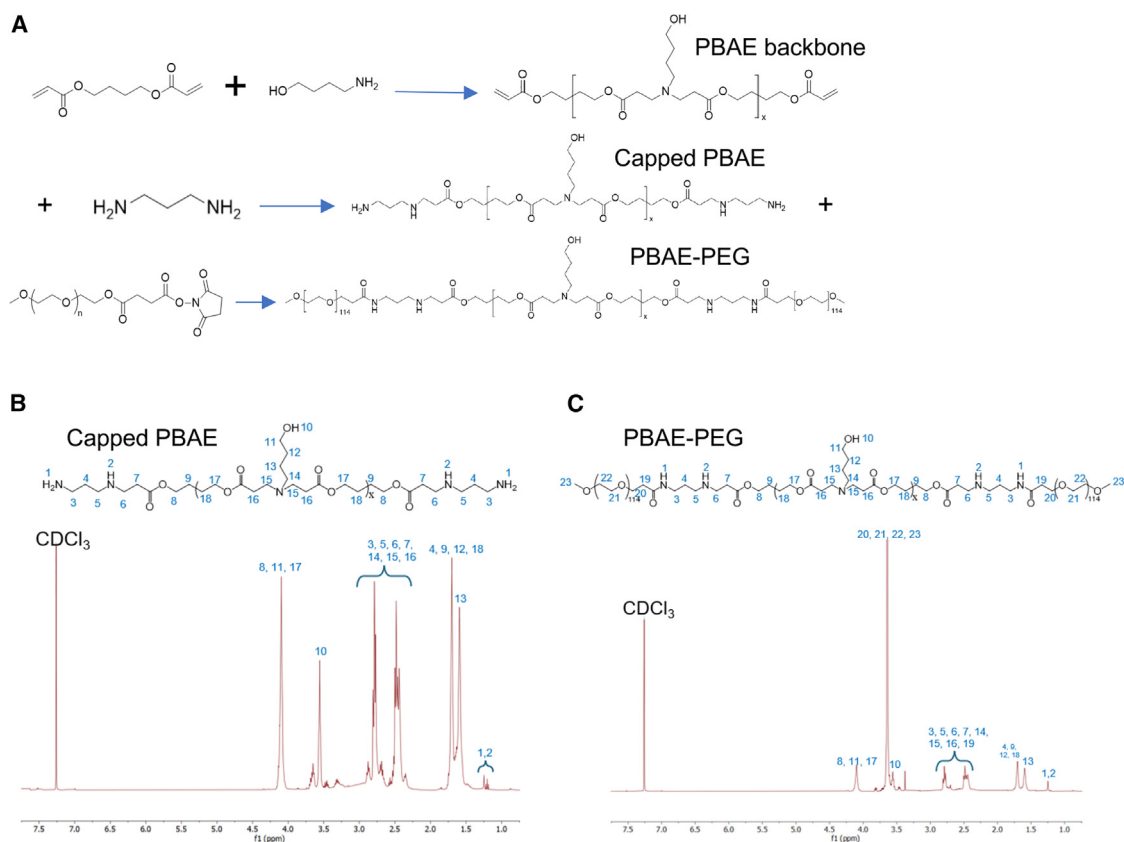
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**Figure 1. Creation of PBAE and PEGylated PBAE**

(A) Uncapped poly( $\beta$ -amino ester)s (PBAE) polymers were synthesized by the Michael addition reaction of 1,4-butanediol diacrylate and 4-amino-1-butanol to produce the PBAE backbone. Subsequently, uncapped PBAE polymers were capped with 1,3-diaminopropane (capped PBAE). PEGylated PBAE (PBAE-PEG) was synthesized by conjugating methoxy-PEG-succinimidyl succinate to capped PBAE at a 2:1 PEG:PBAE molar ratio. (B) Chemical structure of capped PBAE (PBAE) was characterized by  $^1\text{H}$  NMR spectroscopy. (C) Chemical structure of PEGylated PBAE (PBAE-PEG) was characterized by  $^1\text{H}$  NMR spectroscopy.

selectively targeted pulmonary epithelial cells, including alveolar type II (ATII) epithelial cells, while PBAE/LNP were more selective for delivery into airway club epithelial cells.

## RESULTS

### Synthesis of PBAE and PBAE-PEG

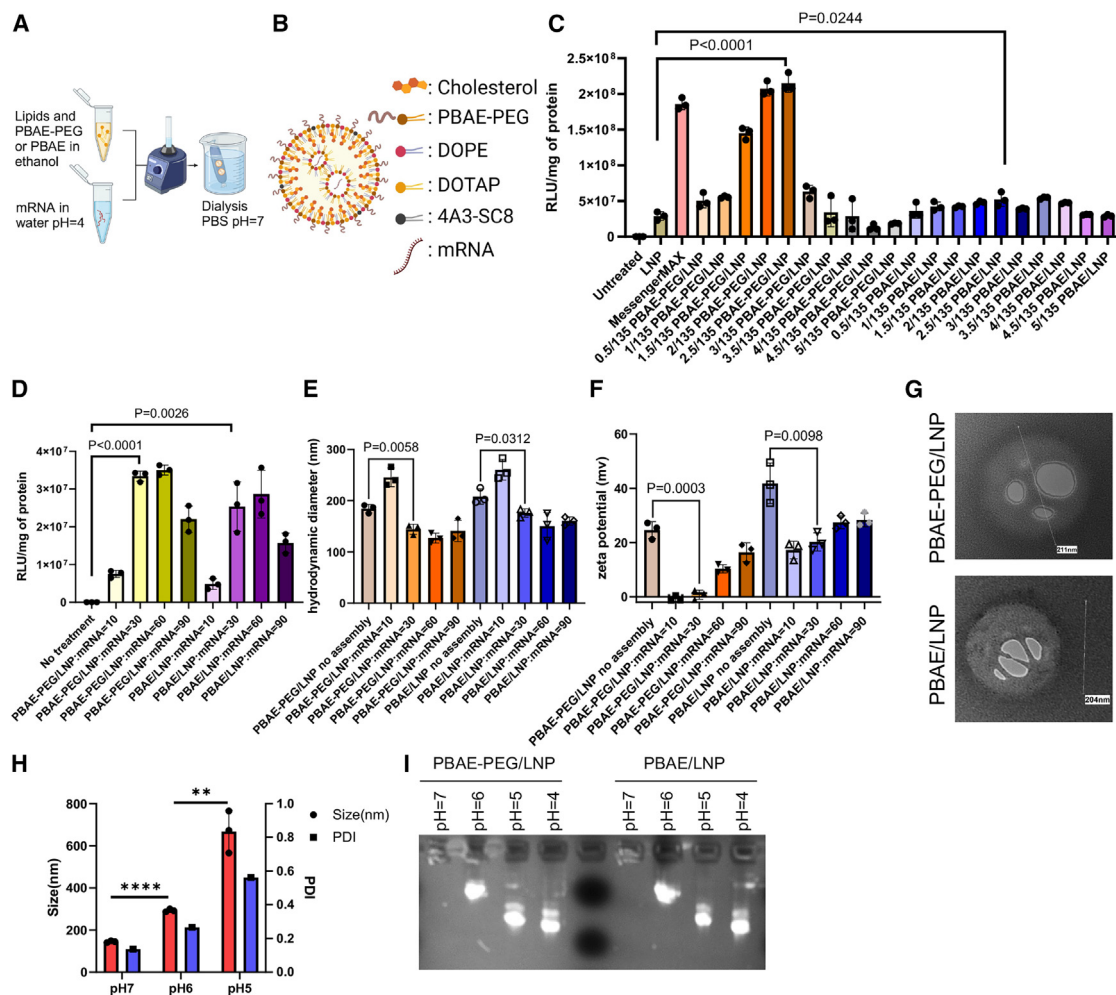
Synthesis of PBAE- or PBAE-PEG-based LNPs was accomplished using methods and techniques that were previously reported by Mastorakos et al.<sup>17</sup> Briefly, PBAE was synthesized through step-growth polymerization via Michael addition to facilitate the reaction between 1,4-butanediol diacrylate and 4-amino-1-butanol. PBAE was then capped by applying hydrophilic 1,3-diaminopropane capping agent. As the PBAE backbone is relatively hydrophobic, this resulted in the creation of amphipathic polymers (Figure 1A). Finally, PBAE-PEG was made by reacting the capped terminal primary amine groups of the PBAE polymer chain with methoxy-PEG-succinimidyl succinates (Figure 1A). Reaction completion was verified by  $^1\text{H}$  NMR (Figures 1B and 1C). For capped PBAE, the chemical shift  $\delta = 1.59$  ppm corresponds to the resonance of the methylene protons beta to the nitrogen in the PBAE backbone, while the chemical shifts

between 1.19 and 1.25 ppm suggest the presence of protons of 1,3-diaminopropane (Figure 1B). For PBAE-PEG, the chemical shift  $\delta = 3.59$  ppm corresponds to the protons of methylene and methyl groups of PEG, indicating the successful formation of PBAE-PEG. The molecular weight of capped PBAE was determined by end-group analysis using  $^1\text{H}$  NMR spectra (Figure 1C) and calculated to be  $\sim 4,400$  g/mol (Table S1), which is close to the value reported by Mastorakos et al.<sup>17</sup>

### In vitro transfection validation of PBAE and PBAE-PEG-based LNPs

The amphipathic polymers PBAE and PBAE-PEG were combined with other amphipathic lipids (4A3-SC8/DOPE/cholesterol/DOTAP dissolved in ethanol) to form PBAE LNPs, which were vortex mixed with mRNA and then dialyzed in PBS at pH 7 (Figure 2A) to encapsulate mRNA in PBAE/LNP or PBAE-PEG/LNP (Figure 2B).

To determine the best molar ratio between PBAE-PEG (or PBAE) and LNP for transfection efficiency *in vitro*, we tested different molar

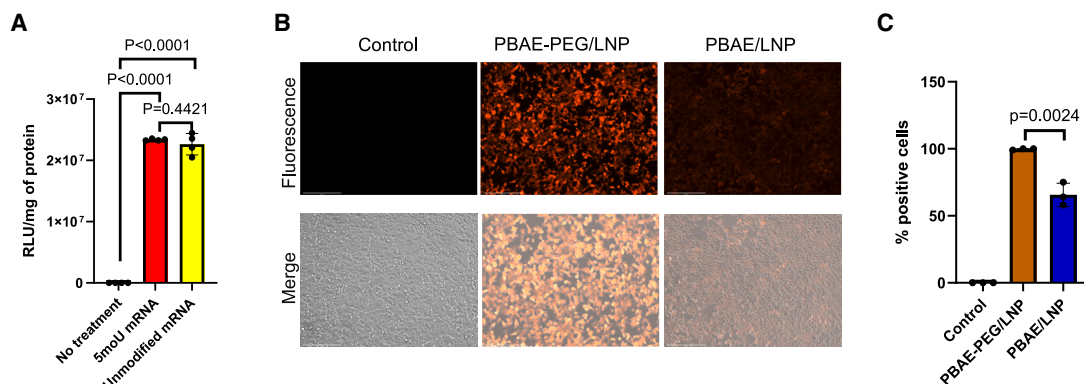


**Figure 2. PBAE-PEG or PBAE-based lipid nanoparticles are capable of transfecting mRNA *in vitro***

(A) PBAE-PEG/lipid nanoparticles (LNPs) or PBAE/LNP dissolved in ethanol were assembled by vortex mixing with mRNA dissolved in water at pH 4 and then dialyzed in PBS at pH 7 overnight. (B) Schematic shows PBAE-PEG/LNP (LNP: 4A3-SC8/DOPE/cholesterol/DOTAP) carrying mRNA. (C) Transfection efficiency of PBAE-PEG/LNP or PBAE/LNP at different molar ratios from 0.5:135 to 5:135 (PBAE-PEG:LNP or PBAE:LNP) carrying unmodified luciferase mRNA (2  $\mu$ g) along with Lipofectamine MessengerMAX was assessed using luciferase assay in HEK293T cells. Forty-eight hours after transfection, luciferase activity was measured. Results are normalized by protein weight and expressed as mean  $\pm$  SD of replicates for each group ( $n = 3$ , unpaired, two-tailed Student's *t* test). (D) Transfection efficiency of PBAE-PEG/LNP or PBAE/LNP with unmodified luciferase mRNA at different mass ratios from 10:1 to 90:1 (PBAE-PEG/LNP:mRNA or PBAE/LNP:mRNA) was assessed using luciferase assay in HEK293T cells as described in (C). (E) Hydrodynamic diameters of PBAE-PEG/LNP or PBAE/LNP with unmodified luciferase mRNA at different mass ratios from 10:1 to 90:1 (PBAE-PEG/LNP:mRNA or PBAE/LNP:mRNA) were measured using Zetasizer. Results are expressed as mean  $\pm$  SD of replicates for each group ( $n = 3$ , unpaired, two-tailed Student's *t* test). (F) The zeta potential of PBAE-PEG/LNP or PBAE/LNP with unmodified luciferase mRNA at different mass ratios were measured as described in (E). Results are also expressed as described in (E). (G) Diameters of PBAE-PEG/LNP (top) and PBAE/LNP (bottom) with unmodified luciferase mRNA (1  $\mu$ g/mL) that compose mesoporous structures were 211 and 204 nm, respectively, which were measured by transmission electron microscopy (TEM). (H) The hydrodynamic diameters of PBAE-PEG/LNP carrying unmodified luciferase mRNA at pH 7, 6, and 5 that were measured using a Zetasizer indicate that PBAE-PEG/LNP with unmodified luciferase mRNA were responsive to pH change. Red and blue bars indicate size (nm) and PDI, respectively. Results are expressed as mean  $\pm$  SD of replicates for each group ( $n = 3$ , unpaired, two-tailed Student's *t* test). \*\*\*\* and \*\* indicate  $p < 0.0001$  and  $p < 0.01$ , respectively. (I) RNA gel electrophoresis indicates the release of unmodified luciferase mRNA from PBAE-PEG/LNP with unmodified luciferase mRNA at a lower pH (pH 4–6) but not at pH 7. Several panels were created with BioRender.com.

ratios ranging from 0.5:135 to 5:135 (PBAE-PEG:LNP or PBAE:LNP) along with only LNP and Lipofectamine MessengerMAX (Thermo Fisher) using luciferase mRNA (unmodified) at a mass ratio of 30:1 (PBAE-PEG/LNP:mRNA or PBAE/LNP:mRNA) in HEK293T cells *in vitro* and assessed transfection efficiency by measuring luciferase

activity. A molar ratio of 2.5:135 provided the highest transfection efficiency for both PBAE-PEG/LNP and PBAE/LNP (Figure 2C). PBAE-PEG/LNP transfection efficiency was similar to Lipofectamine MessengerMAX, followed by PBAE/LNP and LNP alone. Thus, we used a molar ratio of 2.5:135 for subsequent studies.



**Figure 3. PBAE-PEG/LNP and PBAE/LNP can transfect 5moU mRNA *in vitro***

(A) Transfection efficiency of PBAE-PEG/LNP carrying 5moU or unmodified luciferase mRNA (2  $\mu$ g) at the mass ratio of 30:1 (PBAE-PEG/LNP:mRNA) was assessed using luciferase assay in HEK293T cells. Forty-eight hours after transfection, luciferase activity was measured. Results are normalized by protein weight and expressed as mean  $\pm$  SD of replicates for each group ( $n = 3$ , unpaired, two-tailed Student's *t* test). (B) Fluorescence microscopy was conducted to assess transfection efficiency by PBAE-PEG/LNP or PBAE/LNP carrying 5moU mCherry mRNA (2  $\mu$ g) using HEK293T cells. Immunofluorescence images were captured 48 h after transfection. Scale bar indicates 275  $\mu$ m. (C) Transfection efficiency for (B) was quantified by counting fluorescence-positive cells. Results are expressed as mean  $\pm$  SD of replicates for each group ( $n = 3$ , unpaired, two-tailed Student's *t* test).

Next, we sought to determine which mass ratio between PBAE-PEG/LNP (or PBAE/LNP) and mRNA provides the best transfection efficiency using luciferase mRNA in HEK293T cells *in vitro*. Ratios of 30:1 and 60:1 (PBAE-PEG/LNP:mRNA or PBAE/LNP:mRNA) provided the highest transfection efficiency as measured by luciferase activity compared to those of 10:1 or 90:1 (Figure 2D). The size and surface charge of PBAE-PEG/LNP or PBAE/LNP with different amounts of luciferase mRNA (unmodified) was determined by Zetasizer. The size of PBAE-PEG/LNP itself was 184.3 nm. When assembled with the mRNA at a mass ratio of 10:1 (PBAE-PEG/LNP:mRNA), the size significantly increased to 245.6 nm. When assembled with mRNA at a mass ratio of 30:1 or higher (PBAE-PEG/LNP:mRNA), the size significantly decreased to below 144.3 nm (Figure 2E). Likewise, the size of PBAE/LNP alone was 208 nm. When assembled with mRNA at a mass ratio of 10:1 (PBAE/LNP:mRNA), the size significantly increased to 260.3 nm. When assembled with mRNA at a mass ratio of 30:1 or higher (PBAE/LNP:mRNA), the size significantly decreased to below 176 nm (Figure 2E). The zeta potential of PBAE-PEG/LNP or PBAE/LNP itself was 24.6 or 41.8 mV, respectively. When assembled with mRNA at a mass ratio of 30:1, the zeta potential of PBAE-PEG/LNP/mRNA or PBAE/LNP/mRNA was 0.77 and 20.2 mV, respectively. When assembled with mRNA at a mass ratio of 60:1 and 90:1, the zeta potential of PBAE-PEG/LNP/mRNA or PBAE/LNP/mRNA was higher at 10.3 and 27.4 mV, respectively (Figure 2F). These *in vitro* results suggest that PBAE-PEG/LNP or PBAE/LNP with mRNA at the mass ratio of 30:1 or 60:1 is applicable for *in vivo* studies. Due to cost-effectiveness, we chose PBAE-PEG/LNP or PBAE/LNP with mRNA at a mass ratio of 30:1 for the further *in vitro* and *in vivo* studies below.

We assessed the morphological characteristics of PBAE-PEG/LNP or PBAE/LNP with mRNA. Transmission electron microscopy (TEM) indicated that PBAE-PEG/LNP and PBAE/LNP demonstrate a mesoporous structure with sizes of 211 and 204 nm, respectively (Figure 2G).

To predict how PBAE-PEG/LNP releases mRNA after entering cells, we measured the size of PBAE-PEG/LNP in PBS at pH 7 and 100 or 10  $\mu$ M citric acid buffer at pH 6 and pH 5. At neutral pH 7, PBAE-PEG/LNP exhibited an average size of 145.2 nm (polydispersity index [PDI] = 0.137). As the pH decreased, the size increased progressively from 293.0 nm (PDI = 0.266) at pH 6 to 667.9 nm (PDI = 0.562) at pH 5 (Figure 2H), indicating that PBAE-PEG/LNP is responsive to pH change. Consistent with these size distribution results, mRNA was released from PBAE-PEG/LNP at a lower pH (pH 4–6) but not at pH 7 (Figure 2I). These results suggest that PBAE-PEG/LNP releases mRNA in the endocytosis process at low pH.

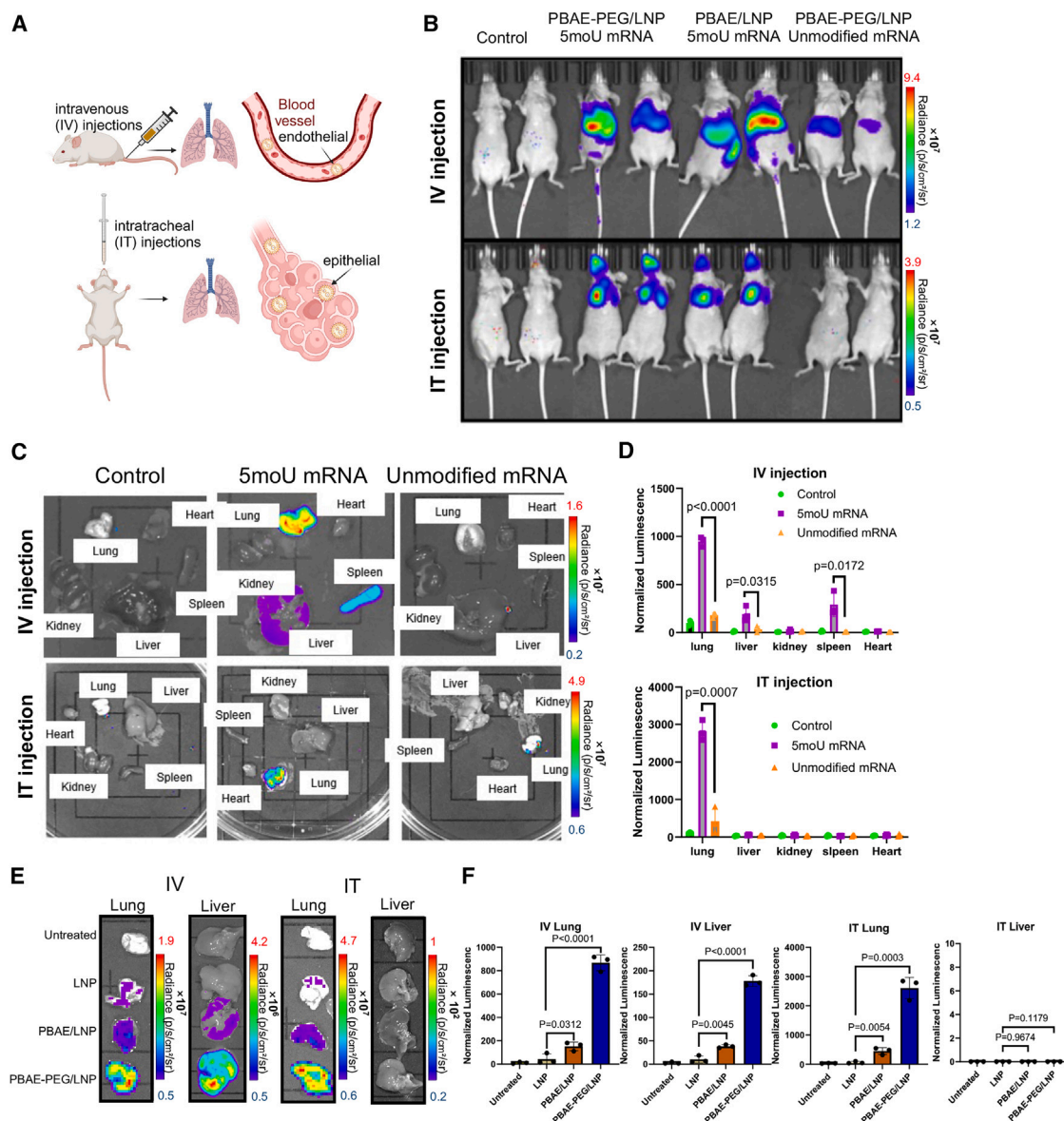
#### ***In vitro* transfection validation of PBAE-PEG/LNP and PBAE/LNP with 5-methoxyuridine modified mRNA**

To be applicable for *in vivo* studies, since 5-methoxyuridine modified (5moU) mRNA is less immunogenic than unmodified mRNA,<sup>21</sup> we tested PBAE-PEG/LNP encapsulating 5moU luciferase mRNA along with unmodified mRNA in HEK293T cells *in vitro*. PBAE-PEG/LNP with 5moU luciferase mRNA significantly induced luciferase activity similar to that achieved with unmodified mRNA (Figure 3A), indicating that PBAE-PEG/LNP can deliver 5moU mRNA as well as unmodified mRNA. The transfection efficiency of PBAE-PEG/LNP or PBAE/LNP with 5moU mRNA was also confirmed by fluorescence analysis using 5moU mCherry mRNA in HEK293T cells *in vitro*, which further indicated that PBAE-PEG/LNP is superior to PBAE/LNP in transfecting HEK293T cells (Figures 3B and 3C).

#### ***In vivo* transfection validation of PBAE-PEG/LNP or PBAE/LNP**

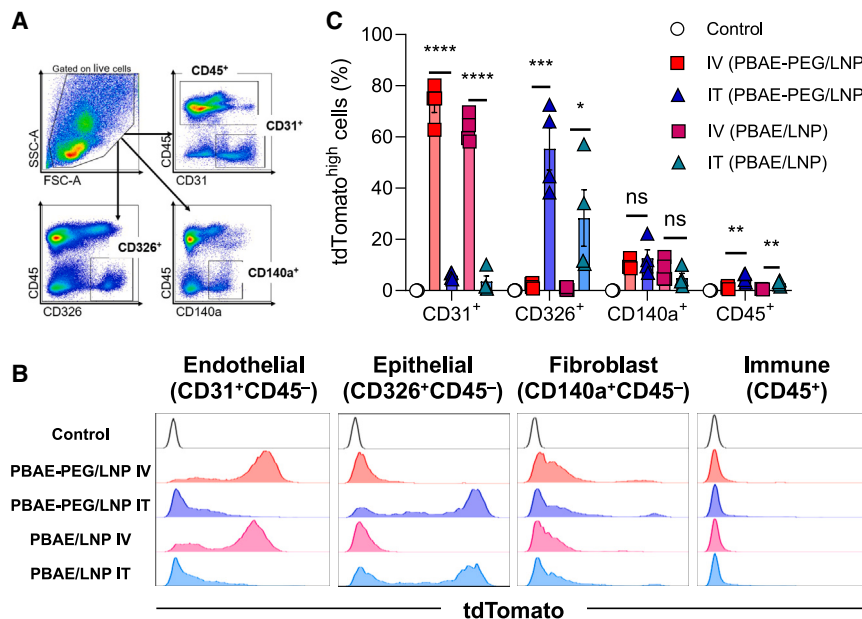
To determine whether PBAE-PEG/LNP or PBAE/LNP can deliver mRNA into lung cells *in vivo*, we compared the transfection efficiency of 5moU or unmodified luciferase mRNA (12.5  $\mu$ g) by PBAE-PEG/LNP at the molar ratio of 2.5:135 using IV or IT injection into NU/J nude mice (Figure 4A). Luciferase activity in a whole mouse body





**Figure 4. PBAE-PEG or PBAE-based LNPs are capable of transfecting mRNA into lung cells *in vivo***

(A) Different routes of injection (intravenous [IV; top] vs. intratracheal [IT; bottom]) with PBAE-PEG/LNP or PBAE/LNP with 5moU luciferase mRNA were conducted to deliver mRNA to the mouse lung. (The drawing was created with [BioRender.com](https://www.biorender.com).) (B) Luminescence from whole bodies of NU/J nude mice ( $n = 4$ ) that were transfected by IV (top) or IT (bottom) injections of PBAE-PEG/LNP with 5moU luciferase mRNA (12.5  $\mu\text{g}$ ), PBAE/LNP with 5moU luciferase mRNA (12.5  $\mu\text{g}$ ), or PBAE-PEG/LNP with unmodified luciferase mRNA (12.5  $\mu\text{g}$ ) were visualized 24 h after transfection using IVIS. The molar ratio of 2.5:135 (PBAE-PEG:LNP or PBAE:LNP) was used. Shown are two representative images of four mice (see [Figure S1](#) for the complete group). (C) Luminescence from different organs (lung, spleen, liver, heart, and kidney) of FVB/N mice ( $n = 3/\text{group}$ ) transfected by IV (top) or IT (bottom) injections of PBAE-PEG/LNP with 5moU or unmodified luciferase mRNA (12.5  $\mu\text{g}$ ) was captured 24 h after transfection using IVIS. The molar ratio of 2.5:135 (PBAE-PEG:LNP) was used. Shown are representative images of one mouse (see [Figure S2](#) for three mice at different PBAE-PEG:LNP or PBAE:LNP molar ratios). (D) Luminescence intensity ( $n = 3/\text{group}$ ) that is described in (C) was quantified by ImageJ. Results are expressed as mean  $\pm$  SD of triplicates for each group (unpaired, two-tailed Student's *t* test). (E) Luminescence from lungs and livers of FVB/N mice ( $n = 3/\text{group}$ ) transfected by IV (left) or IT (right) injections of only LNP, PBAE/LNP, or PBAE-PEG/LNP with 5moU luciferase mRNA (10  $\mu\text{g}$ ) was captured 12 h after transfection using IVIS. A molar ratio of 2.5:135 (PBAE-PEG:LNP or PBAE:LNP) was used. (F) Luminescence intensity from lungs and livers described in (E) was quantified by ImageJ. Results are expressed as mean  $\pm$  SD of triplicates for each group (unpaired, two-tailed Student's *t* test).



**Figure 5. PBAAE-PEG/LNP and PBAAE/LNP deliver mRNA into lung endothelial cells by IV injection and lung epithelial cells by IT injection**

(A) Gating strategy of fluorescence-activated cell sorting (FACS) analysis for the identification of cell lineage populations (CD31<sup>+</sup>CD45<sup>-</sup> as endothelial cells, CD326<sup>+</sup>CD45<sup>-</sup> as epithelial cells, CD140a<sup>+</sup>CD45<sup>-</sup> as fibroblasts, and CD45<sup>+</sup> as immune cells) from whole-lung single-cell suspension is shown. (B) Representative FACS histograms show tdTomato expression in endothelial cells (CD31<sup>+</sup>CD45<sup>-</sup>), epithelial cells (CD326<sup>+</sup>CD45<sup>-</sup>), lung fibroblasts (CD140a<sup>+</sup>CD45<sup>-</sup>), and immune cells (CD45<sup>+</sup>) from lungs of Ai14 tdTomato mice 4 days after IV or IT injection of PBAAE-PEG/LNP and PBAAE/LNP carrying 5moU Cre mRNA (12.5 μg administered twice on days 0 and 2). (C) Graph displays the frequency of tdTomato<sup>high</sup> cells in endothelial cells, epithelial cells, fibroblasts, and immune cells obtained as described in (B). Results were pooled from three independent experiments with four mice per group. Each symbol represents an individual mouse. Results are expressed as mean ± SD of replicates for each group ( $n = 4$ , unpaired, two-tailed Student's  $t$  test). \*\*\*\*, \*\*\*, \*\*, and \* indicate  $p < 0.0001$ ,  $p < 0.001$ ,  $p < 0.01$ , and  $p < 0.05$ , respectively.

assessed by *in vivo* imaging systems (IVIS) indicated that 5moU luciferase mRNA was superior to unmodified luciferase mRNA for transfection into multiple organs using PBAAE-PEG/LNP or PBAAE/LNP *in vivo* regardless of IV or IT injection (Figures 4B and S1). Based on these data, we focused on PBAAE-PEG/LNP or PBAAE/LNP using 5moU mRNA for further *in vivo* studies.

To determine which organs were transfected by PBAAE-PEG/LNP or PBAAE/LNP with 5moU luciferase mRNA, we extracted five organs (lung, liver, spleen, kidney, and heart) from FVB/N mice that were transfected with PBAAE-PEG/LNP or PBAAE/LNP at different molar ratios of PBAAE-PEG (or PBAAE):LNP (1:135, 2.5:135, and 4:135) carrying 5moU luciferase mRNA (10 μg) by IV or IT. Regardless of molar ratio, IV-injected PBAAE-PEG/LNP with 5moU luciferase mRNA provided the highest luciferase activity in lungs, followed by spleens and livers. However, IT-injected PBAAE-PEG/LNP with 5moU luciferase mRNA provided luciferase activity in lungs but not in other organs regardless of molar ratio (Figures 4C, 4D, and S2A). We quantified which molar ratio of PBAAE-PEG (or PBAAE):LNP provided the highest lung luciferase activity and found that the 2.5:135 molar ratio was significantly better than the other molar ratios (1:135 and 4:135) for both PBAAE-PEG/LNP and PBAAE/LNP, regardless of route of injection (Figures S2B and S2C). We also compared the *in vivo* transfection efficiency of PBAAE-PEG/LNP or PBAAE/LNP to LNP alone. As shown in Figures 4E and 4F, luciferase activity delivered by PBAAE-PEG/LNP was the highest, followed by PBAAE/LNP and LNP only for IV-mediated lung and liver transfection and IT-mediated lung transfection, indicating that both PBAAE-PEG/LNP and PBAAE/LNP are superior to LNP only for *in vivo* transfection targeting lung and/or liver. Of note, neither PBAAE-PEG/LNP nor PBAAE/LNP with 5moU luciferase mRNA (12.5 μg) administered by IV or IT injection caused liver toxicity

(Figure S3A; no increase in alanine aminotransferase [ALT] was observed) or lung inflammation (Figure S3B).

#### PBAAE-PEG/LNP or PBAAE/LNP deliver mRNA into lung endothelial cells by IV injection and lung epithelial cells by IT injection *in vivo*

To determine which lung cell types are transfected by PBAAE-PEG/LNP or PBAAE/LNP with 5moU mRNA *in vivo*, we performed fluorescence-activated cell sorting (FACS) analysis using isolated Ai14 tdTomato mouse lung cells transfected with 5moU Cre mRNA by IT and IV injection (Figure 5A). In Ai14 tdTomato mice, tdTomato is induced upon *loxP* recombination driven by transfected Cre.<sup>22</sup> IV injection of PBAAE-PEG/LNP or PBAAE/LNP with Cre mRNA resulted in transfection into PECAM-1 (CD31)<sup>+</sup> lung endothelial cells at 73.2% or 63.0% efficiency, respectively, but not into EpCAM (CD326)<sup>+</sup> lung epithelial cells, PDGFRA (CD140a)<sup>+</sup> lung fibroblasts, or PTPRC (CD45)<sup>+</sup> lung immune cells (Figures 5B and 5C). However, IT injection of PBAAE-PEG/LNP or PBAAE/LNP with Cre mRNA resulted in transfection into EpCAM (CD326)<sup>+</sup> lung epithelial cells at 55.4% or 28.3% efficiency, respectively, but not into PECAM-1 (CD31)<sup>+</sup> lung endothelial cells, PDGFRA (CD140a)<sup>+</sup> lung fibroblasts, or PTPRC (CD45)<sup>+</sup> lung immune cells (Figures 5B and 5C). These results indicate that our PBAAE-PEG/LNP and PBAAE/LNP target specific lung cell types (lung endothelial cells or lung epithelial cells) depending on the injection route (IV or IT).

#### PBAAE-PEG/LNP or PBAAE/LNP delivers mRNA into ERG<sup>+</sup> lung endothelial cells but not into lung alveolar epithelial or airway epithelial cells when delivered IV *in vivo*

We also validated lung cell types in which PBAAE-PEG/LNP or PBAAE/LNP containing 5moU Cre mRNA is delivered by IV administration

to Ai14 tdTomato mice using confocal immunofluorescence. Consistent with the FACS analysis (Figure 5), tdTomato expression, which was induced by IV injection of both PBAE-PEG/LNP and PBAE/LNP with 5moU Cre mRNA, was colocalized with ERG (lung endothelial cell marker; see Figures 6A and S4) but not NKX2-1 (lung epithelial cell marker; see Figures 6B and S5), LAMP3 (lung ATII cell marker; see Figures 6B and S5), SCGB1A1 (lung airway club cell marker; see Figures 6C and S6), TUBA1A (lung airway ciliated cell marker; see Figures 6C and S6), or MEOX2 (lung fibroblast marker; see Figures 6D and S7), indicating that IV injection of PBAE-PEG/LNP or PBAE/LNP with 5moU mRNA delivers mRNA into lung endothelial cells specifically.

**PBAE-PEG/LNP delivers mRNA into both lung ATII epithelial cells and lung airway club epithelial cells, while PBAE/LNP delivers mRNA into lung airway club epithelial cells when delivered IT *in vivo***

FACS data indicated that IT injection of PBAE-PEG/LNP or PBAE/LNP with 5moU mRNA delivers mRNA into EpCAM (CD326)<sup>+</sup> lung epithelial cells (Figure 5). We used confocal immunofluorescence to determine which lung epithelial cell types are transfected with 5moU Cre mRNA in Ai14 tdTomato mice by IT injection. While PBAE-PEG/LNP did not induce tdTomato expression in ERG<sup>+</sup> lung endothelial cells (Figures 7A and S8), PBAE-PEG/LNP induced tdTomato expression in both LAMP3/NKX2-1<sup>+</sup> lung ATII epithelial cells (Figures 7B, S9, and S10) and SCGB1A1<sup>+</sup> lung airway club epithelial cells (Figures 7C, S10, and S11). PBAE-PEG/LNP did not induce tdTomato expression in TUBA1A<sup>+</sup> lung airway ciliated epithelial cells (Figures 7C and S11) or MEOX2<sup>+</sup> lung fibroblasts (Figures 7D and S12).

While PBAE/LNP did not induce tdTomato expression in ERG<sup>+</sup> lung endothelial cells either (Figures 7A and S8), PBAE/LNP induced tdTomato expression significantly more in SCGB1A1<sup>+</sup> lung airway club epithelial cells (Figures 7C, S10, and S11) than in LAMP3/NKX2-1<sup>+</sup> lung ATII epithelial cells (Figures 7B, S9, and S10). PBAE/LNP did not induce tdTomato expression in TUBA1A<sup>+</sup> lung airway ciliated epithelial cells (Figures 7C and S11) or MEOX2<sup>+</sup> lung fibroblasts (Figures 7D and S12). These data indicate that PBAE-PEG/LNP administered by *in vivo* IT injection is superior for the targeting of both lung ATII epithelial cells and airway club epithelial cells, while PBAE/LNP is superior for specifically targeting lung airway club epithelial cells over lung ATII epithelial cells.

## DISCUSSION

Despite advances in nanoparticle technology, pulmonary cell-specific delivery systems remain challenging and have not been clinically applied yet. In our study, we developed two nanoparticles, PBAE-PEG/LNP and PBAE/LNP, to deliver mRNA into lung endothelial cells via IV injection and into lung epithelial cells, including lung ATII cells and airway club cells, via IT injection without causing liver toxicity and lung inflammation. Our results demonstrated that PEGylation affects targeting ability. Specifically, PBAE-PEG/LNP delivered mRNA into lung epithelial cells, including ATII and airway

club cells, while PBAE/LNP targeted airway club cells better than ATII cells via IT injection.

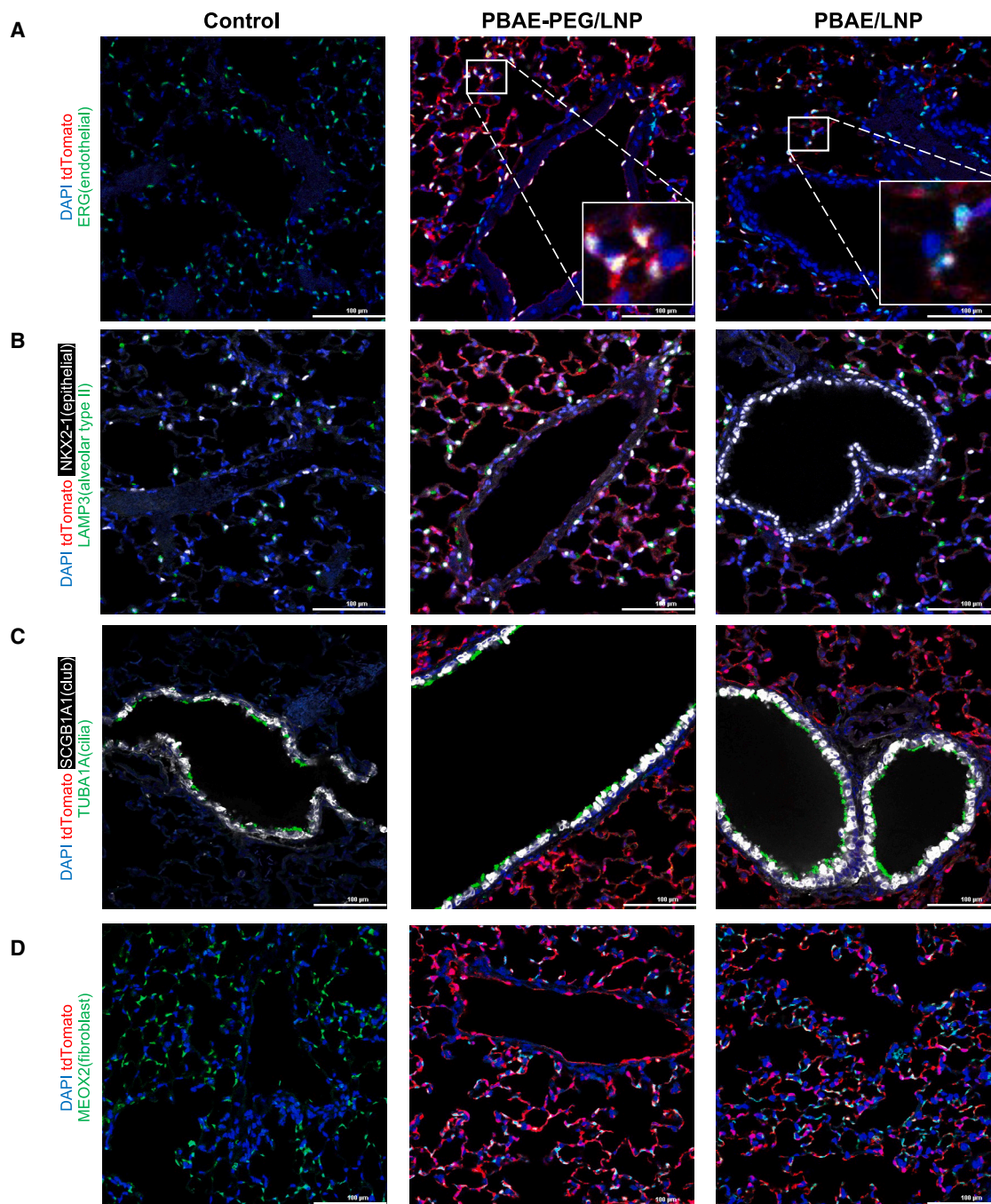
In this study, PEGylation of PBAE facilitated delivery of mRNA into lung ATII epithelial cells beyond airway club epithelial cells. We hypothesize that PEGylation helps to (1) stabilize LNPs, (2) reduce their immunogenicity, and (3) prevent phagocytosis by alveolar macrophages, which results in wider targeting of lung epithelial cells in the alveolar regions. PBAE-PEG/LNP can be considered a potential candidate for delivering therapeutic cargo into ATII epithelial cells for genetic disorders, for example, ABCA3 deficiency. However, PBAE/LNP also presents a promising alternative for specific pulmonary conditions where targeted gene delivery into airway club epithelial cells, which serve as progenitors of goblet and ciliated epithelial cells, is needed (e.g., cystic fibrosis).

We have designed a formulation of LNPs for mRNA delivery based on previously reported LNP formulations.<sup>10,15,17</sup> However, as reported previously, Dlin-MC3-DMA and SM-102 can be used instead of 4A3-SC8 for ionizable lipids.<sup>12,23</sup> Different combinations of helper lipids, including DOPE, distearoylphosphatidylcholine, dimyristoyl glycerol-PEG, and C14-PEG, may improve the transfection efficiency.<sup>10,15</sup> For further optimization, probably the most efficient approach would be to combine machine learning with high-throughput lipid synthesis,<sup>24</sup> which might provide the best LNP formulation for IT and/or IV administration of mRNA in the future.

For the COVID-19 vaccine, N1-methylpseudouridine-modified mRNA was used to reduce mRNA immunogenicity and increase the half-life of synthetic mRNA.<sup>25,26</sup> In our study, we used 5moU mRNA commercially available from TriLink, which provided successful transfection efficiency with PBAE-PEG/LNP and PBAE/LNP *in vivo*. However, N1-methylpseudouridine-modified mRNA might provide better transfection efficiency considering that protein expression from N1-methylpseudouridine-modified mRNA is higher than that from 5moU mRNA and the immunogenicity from N1-methylpseudouridine-modified mRNA is lower than that from 5moU mRNA according to the products at the TriLink Biotechnologies site (<https://www.trilinkbiotech.com/browse/stocked-mrnas>). We will test N1-methylpseudouridine-modified mRNA with PBAE-PEG/LNP and PBAE/LNP in a future study.

In summary, our work proposes that PBAE-PEG/LNP and PBAE/LNP have different targeting abilities after IT or IV injection. Our study demonstrated the possibility of selective gene transfer for future therapeutic mRNA or genome editing via Cas9 mRNA and guide RNA(s) into pulmonary endothelial and epithelial cells. Our study provides a framework for designing temporary or permanent strategies for pulmonary gene therapy for disorders caused by mutations in ABCA3, CFTR, SFTPB, and SFTPC, and other monogenic lung diseases. Further studies are needed to assess the duration of expression and whether levels of protein expression will be sufficient to be an effective treatment to genetic disorders.

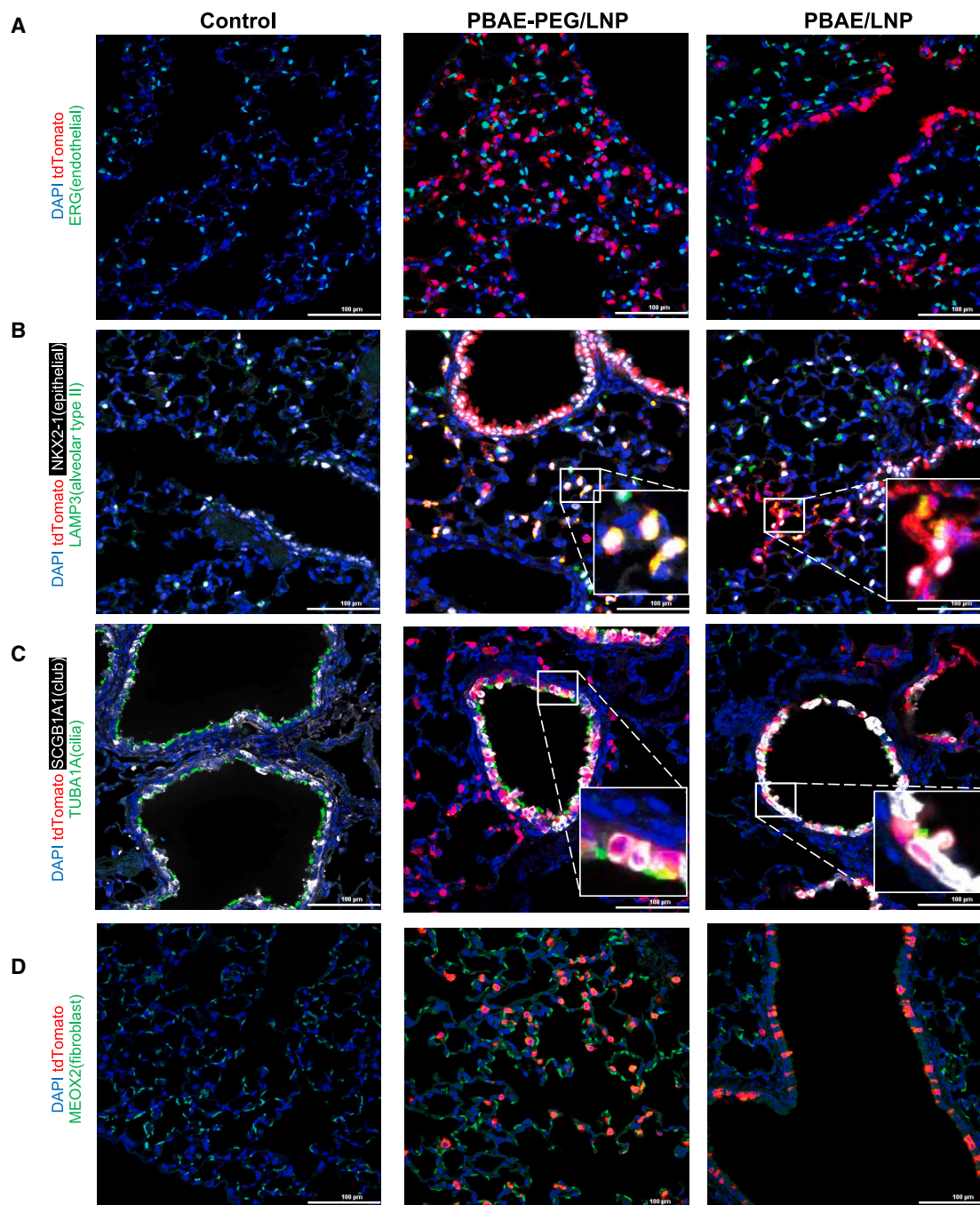




**Figure 6. IV injection of PBAE-PEG/LNP and PBAE/LNP carrying mRNA targets  $ERG^+$  lung endothelial cells**

Lungs of Ai14 tdTomato mice 4 days after transfection with PBAE-PEG/LNP and PBAE/LNP carrying 5moU Cre mRNA were harvested and sectioned for immunofluorescence staining to identify which lung cell types were transfected with Cre mRNA (12.5 μg administered twice on days 0 and 2), resulting in tdTomato expression as described in Figure 5. Cell nuclei were counterstained with DAPI (blue). Shown are representative images from three mice. Scale bar indicates 100 μm. (A) tdTomato (red) was co-expressed with the endothelial cell marker ERG (green). (B) tdTomato (red) was co-expressed with neither the lung epithelial cell marker NKX2-1 (white) nor the alveolar type II (ATII) epithelial cell marker LAMP3 (green). (C) tdTomato (red) was co-expressed with neither the airway club epithelial cell marker SCGB1A1 (white) nor the airway ciliated epithelial cell marker TUBA1A (green). (D) tdTomato (red) was not co-expressed with the fibroblast marker MEOX2 (green).





**Figure 7. IT injection of PBAE-PEG/LNP with mRNA targets both NKX2-1/LAMP3<sup>+</sup> ATII epithelial cells and SCGB1A1<sup>+</sup> airway club epithelial cells, while PBAE/LNP with mRNA targets SCGB1A1<sup>+</sup> airway club epithelial cells better than NKX2-1/LAMP3<sup>+</sup> ATII epithelial cells in mouse lungs**

Lungs of Ai14 tdTomato mice 4 days after transfection with PBAE-PEG/LNP and PBAE/LNP carrying 5moU Cre mRNA were harvested and sectioned for immunofluorescence staining to identify which lung cell types were transfected with Cre mRNA (12.5  $\mu$ g administered twice on days 0 and 2), resulting in tdTomato expression as described in Figure 5. Cell nuclei were counterstained with DAPI (blue). Shown are representative images from three mice. Scale bar indicates 100  $\mu$ m. (A) tdTomato (red) was not co-expressed with the endothelial cell marker ERG (green). (B) tdTomato (red) was co-expressed with the lung epithelial cell marker NKX2-1 (white) and the ATII epithelial cell marker LAMP3 (green). (C) tdTomato (red) was co-expressed with the airway club epithelial cell marker SCGB1A1 (white) but not the airway ciliated epithelial cell marker TUBA1A (green). (D) tdTomato (red) was not co-expressed with the fibroblast marker MEOX2 (green).

## MATERIALS AND METHODS

### Materials

We purchased 1,4-butanediol diacrylate (catalog no. 411744), 4-amino-1-butanol (catalog no. 178330), 1,3-diaminopropane (catalog no. D23602), citric acid (catalog no. C0759), and ALT Activity Assay (catalog no. MAK052) from Sigma-Aldrich (St. Louis, MO). We purchased 18:1 TAP (DOTAP) (catalog no. 890890) and 18:1 DOPE (catalog no. 850725) from Avanti Polar Lipids (Alabaster, AL). We purchased 4A3-SC8 (catalog no. N-1438) from Echelon Biosciences (Salt Lake City, UT). Methoxy PEG Succinimidyl Succinate (catalog no. M-SS-5000) was purchased from JenKem Technology USA (Plano, TX). CleanCap mCherry mRNA (5moU) (catalog no. L-7203), CleanCap FLuc mRNA (5moU) (catalog no. L-7202), and CleanCap Cre mRNA (catalog no. L-7211) were purchased from TriLink Biotechnologies (San Diego, CA). D-Luciferin Bioluminescent Substrate (catalog no. 770504) was purchased from Revvity (Waltham, MA).

### Synthesis of PBAE and PEGylated PBAE

Synthesis of the PBAE polymer backbone was conducted via a two-step Michael addition reaction, where 1,4-butanediol diacrylate and 4-amino-1-butanol were combined in a 1.2:1 molar ratio and reacted at 90°C for 24 h to produce uncapped PBAE polymers. These polymers were subsequently precipitated three times in cold ether to eliminate any unreacted materials and then dried under vacuum. For the capping of the terminal acrylate groups, the PBAE polymers were solubilized in tetrahydrofuran (THF) at a concentration of 100 µg/mL, and 30 molar equiv of 1,3-diaminopropane were introduced. The reaction was carried out with stirring at room temperature for 4 to 5 h. Finally, the end-capped PBAE polymers were isolated by precipitation in cold ether to remove any unreacted reagents and then dried under vacuum. Methoxy-PEG-succinimidyl succinates were then conjugated to the two terminal primary amine groups on both ends of the 1,3-diaminopropane-capped PBAE polymer chain. This process involved transferring the capped PBAE polymers along with 2.05 molar equiv of 5 kDa methoxy-PEG-succinimidyl succinate into a glass vial, which was then evacuated and filled with nitrogen. The reactants were dissolved in THF, and the mixture was stirred at room temperature overnight. The resultant PBAE-PEG polymer was then precipitated and washed three times with cold ether, followed by drying under vacuum. PBAE polymers were stored at −20°C.

### Creation of PBAE-PEG/LNP and PBAE/LNP

LNP without PBAE-PEG or PBAE was prepared by dissolving 4A3-SC8, DOPE, cholesterol, and DOTAP at a molar ratio of 20:20:35:60 in ethanol. PBAE-PEG/LNP and PBAE/LNP were prepared by dissolving lipids (4A3-SC8, DOPE, cholesterol, and DOTAP) along with PBAE-PEG or PBAE in ethanol at the molar ratio of 20:20:35:60:2.5. Concurrently, mRNA was dissolved in a 10 mM citrate buffer at pH 4. The lipids in ethanol and the RNA in the citrate buffer were then rapidly mixed using a vortex as described previously<sup>10</sup> at an ethanol-aqueous volume ratio of 1:3. This mixture

was incubated at room temperature for 10 min and then dialyzed in PBS at pH 7 overnight, which achieved a final weight ratio of 30:1 (total lipids:mRNA). The PBAE-PEG/LNP formulation comprises an LNP composition that includes 4A3-SC8, DOPE, cholesterol, PBAE-PEG, and DOTAP in a molar ratio of 20:20:35:2.5:60, respectively. The PBAE/LNP formulation consists of 4A3-SC8, DOPE, cholesterol, PBAE, and DOTAP, also in a molar ratio of 20:20:35:2.5:60.

### *In vitro* HEK293T cell transfections

The HEK293T cell line was cultured in Dulbecco's Modified Eagle's Medium (catalog no. 11960044; Thermo Fisher, Waltham, MA) enriched with 10% fetal bovine serum (FBS) (catalog no. F4135; Sigma-Aldrich) and 1% penicillin/streptomycin (catalog no. 15140122; Thermo Fisher). Cell cultures were incubated at 37°C and 5% CO<sub>2</sub>. A total of 2 µg (each well of a 24-well plate) of mCherry 5moU (or unmodified) mRNA or luciferase 5moU (or unmodified) mRNA was transfected using PBAE-PEG/LNP, PBAE/LNP at different molar ratios (0:135, 0.5:135, 1:135, 1.5:135, 2:135, 2.5:135, 3:135, 3.5:135, 4:135, 4.5:135, and 5:135), or Lipofectamine MessengerMAX into HEK293T cells, which were harvested 24–48 h after transfection. Luciferase assays were performed in accordance with the Luciferase Assay System (catalog no. E1501; Promega, Madison, WI). Fluorescence images were obtained with an EVOS fluorescence microscope (Thermo Fisher).

### Characterization of PBAE-PEG/LNP and PBAE/LNP

The hydrodynamic diameter and surface potential of PBAE-PEG/LNP and PBAE/LNP were measured using Zetasizer (Malvern, Malvern, UK). Chemical structures of polymer products with chloroform-D (CDCl<sub>3</sub>) as the solvent were characterized by <sup>1</sup>H NMR spectroscopy using a Bruker AV 400 MHz spectrometer (Billerica, MA). The molecular weight of capped PBAE was determined by end-group analysis of the <sup>1</sup>H NMR result.<sup>27</sup> TEM on PBAE-PEG/LNP and PBAE/LNP that carried unmodified luciferase mRNA at a concentration of 1 µg/mL was conducted by the Integrated Pathology Research Facility at Cincinnati Children's Hospital Medical Center (CCHMC). Gel electrophoresis was performed using a MOPS-based 1.2% agarose gel containing formaldehyde and ethidium bromide in 1× MOPS buffer.

### Mouse studies

All animal experiments conducted were approved by the Institutional Animal Care and Use Committee. For *in vivo* whole-body imaging studies with PBAE-PEG/LNP or PBAE/LNP encapsulating 5moU or unmodified luciferase mRNA, NU/J hairless nude mice aged 8–12 weeks obtained from The Jackson Laboratory were used. At 24 h after IV or IT injection, mice received an intraperitoneal injection of 200 µL D-luciferin (30 mg/mL). To assess *in vivo* luciferase activity, we used IVIS Lumina III and Spectrum CT *in vivo* imaging systems (PerkinElmer, Waltham, MA). Whole-body imaging was performed 10 min after D-luciferin administration, and luminescence intensity was quantified using Living Image software (Spectral Instruments, Tucson, AZ). For *ex vivo* imaging, FVB/N mice aged 8–12 weeks

obtained from The Jackson Laboratory were administered PBAE-PEG/LNP or PBAE/LNP encapsulating 5moU or unmodified luciferase mRNA by IV or IT injection. At 12 or 24 h after injection, mice received an intraperitoneal injection of 200  $\mu$ L D-luciferin (30 mg/mL) and were euthanized 10 min later. Internal organs such as heart, liver, spleen, lung, and kidney were collected for luminescence imaging by IVIS Lumina III and Spectrum CT *in vivo* imaging systems (PerkinElmer). For the cell-specific transfection study, B6.Cg-Gt(ROSA)26Sor<sup>tm14(CAG-tdTomato)Hze</sup>/J(Ai14) Cre reporter mice (Ai14 tdTomato mice) were obtained from The Jackson Laboratory. PBAE-PEG/LNP or PBAE/LNP with 5moU Cre mRNA were administered to Ai14 tdTomato mice by either IV or IT injection. Lungs were harvested 4 days after the PBAE-PEG/LNP or PBAE/LNP administration containing mRNA. Single-cell suspensions were made from the mouse lungs. Lung sections were also prepared from the mouse lungs for lung histology analysis using hematoxylin and eosin staining. Plasma ALT activity, an indicator of liver injury, was measured using a Diagnostic ALT test kit from Sigma-Aldrich (catalog no. MAK052) according to the manufacturer's instructions.

### Flow cytometry analysis

Lungs from Ai14 tdTomato mice were digested in PBS medium with 480 U/mL collagenase type 1 (catalog no. 17100017; Thermo Fisher), 5 U/mL dispase (catalog no. CLS354235; Sigma-Aldrich), and 0.025 mg/mL DNase I (catalog no. D-300-100; Gold Biotechnology, St. Louis, MO). Red blood cells (RBCs) were removed using RBC lysis buffer (catalog no. 420301; BioLegend, San Diego, CA) for 5 min on ice. Single-cell suspensions from mouse lungs ( $2 \times 10^6$  cells/well, 96-well plate) were stained with an antibody cocktail for 30 min on ice. The following antibodies were used for surface staining: CD45 (clone: I3/2.3, BioLegend), CD31 (clone: 390, Invitrogen, Waltham, MA), CD326 (clone: G8.8, Invitrogen), and CD140a (clone: APA5, Invitrogen). To exclude dead cells from the analysis, the LIVE/DEAD Fixable Near-IR Dead Cell Stain Kit (Invitrogen) was used. After surface staining, cells were washed three times with PBS containing 2% fetal bovine serum (catalog no. A5256701; Thermo Fisher). Samples were run on a BD LSRFortessa (BD Biosciences, Franklin Lakes, NJ), and data were analyzed using FlowJo (BD Biosciences).

### Confocal immunofluorescence analysis

Lungs harvested from Ai14 tdTomato mice were sectioned at 5  $\mu$ m for confocal immunofluorescence studies using a previously established protocol.<sup>28</sup> Specific cell types within these sections were identified using a series of primary antibodies: 1:200 goat anti-tdTomato (catalog no. AB8181; OriGene, Rockville, MD) to detect tdTomato expression,<sup>29</sup> 1:500 rabbit anti-ERG (catalog no. EPR3864; Abcam, Cambridge, UK) for lung endothelial cells,<sup>30</sup> 1:1,000 rabbit anti-NKX2-1 (catalog no. 8G7G3-1; Seven Hills, Cincinnati, OH) for lung epithelial cells,<sup>31</sup> 1:500 guinea pig anti-LAMP3 (catalog no. 391 005; SYSY, Göttingen, Germany) for ATII epithelial cells,<sup>32</sup> 1:500 rabbit anti-SCGB1A1 (catalog no. WRAB-3950; Seven Hills) for airway club epithelial cells,<sup>33</sup> 1:500 mouse anti-TUBA4A (anti-TUBA1A in mouse; catalog no. T7451; Sigma-Aldrich) for airway ciliated epithelial cells,<sup>34</sup> and 1:50 rabbit anti-MEOX2 (catalog no.

HPA053793; Atlas Antibodies, Bromma, Sweden) for lung fibroblasts.<sup>35</sup> Sections were incubated with labeled secondary antibodies (Invitrogen) to visualize the primary antibody binding: donkey anti-rabbit Alexa Fluor AF 555 (catalog no. A-31572) and AF 647 (catalog no. A32733) for rabbit primary antibodies, goat anti-mouse AF 488 (catalog no. A-11001) for mouse primary antibodies, donkey anti-guinea pig AF 488 (catalog no. A-11073) for guinea pig primary antibodies, and donkey anti-goat AF 555 (catalog no. A-21432) for goat primary antibodies. Fluorescence images were captured using a Nikon A1R confocal microscope. The captured images were analyzed using Imaris software (Nikon, Melville, NY).

### DATA AVAILABILITY

All the raw data is available in Table S2.

### ACKNOWLEDGMENTS

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### AUTHOR CONTRIBUTIONS

Conceptualization, B.L., A.G.Z., D.S., J.A.W., and Y.M. Methodology, B.L., Y.S., N.L., Y.H., W.D.S., A.S., X.C., M.E.S., and Y.M. Investigation, B.L., J.A.W., and Y.M. Data curation, B.L., Y.S., N.L., Y.H., K.A., J.A.W., and Y.M. Formal analysis, Y.M. Validation, Y.M. Writing, B.L., W.D.S., J.A.W., and Y.M. Resources, J.A.W. and Y.M. Supervision, Y.M. Project administration, J.A.W. and Y.M. Funding acquisition, K.A., J.A.W., and Y.M.

### DECLARATION OF INTERESTS

The authors filed an invention disclosure on the combination of PBAE-PEG or PBAE with LNPs.

### SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ymthe.2025.01.037>.

### REFERENCES

1. Noguee, L.M., Dunbar, A.E., Wert, S.E., Askin, F., Hamvas, A., and Whitsett, J.A. (2001). A Mutation in the Surfactant Protein C Gene Associated with Familial Interstitial Lung Disease. *N. Engl. J. Med.* 344, 573–579. <https://doi.org/10.1056/NEJM200102223440805>.
2. Noguee, L.M., Garnier, G., Dietz, H.C., Singer, L., Murphy, A.M., DeMello, D.E., and Colten, H.R. (1994). A mutation in the surfactant protein B gene responsible for fatal neonatal respiratory disease in multiple kindreds. *J. Clin. Invest.* 93, 1860–1863. <https://doi.org/10.1172/JCI117173>.
3. Devriendt, K., Vanhole, C., Matthijs, G., and de Zegher, F. (1998). Deletion of Thyroid Transcription Factor-1 Gene in an Infant with Neonatal Thyroid Dysfunction and Respiratory Failure. *N. Engl. J. Med.* 338, 1317–1318. <https://doi.org/10.1056/NEJM199804303381817>.



4. Shulenin, S., Noguee, L.M., Annilo, T., Wert, S.E., Whitsett, J.A., and Dean, M. (2004). ABCA3 Gene Mutations in Newborns with Fatal Surfactant Deficiency. *N. Engl. J. Med.* 350, 1296–1303. <https://doi.org/10.1056/NEJM0A032178>.
5. Kerem, B., Rommens, J.M., Buchanan, J.A., Markiewicz, D., Cox, T.K., Chakravarti, A., Buchwald, M., and Tsui, L.C. (1989). Identification of the Cystic Fibrosis Gene: Genetic Analysis. *Science* 245, 1073–1080. <https://doi.org/10.1126/SCIENCE.2570460>.
6. Kerstjens-Frederikse, W.S., Bongers, E.M.H.F., Roofthoof, M.T.R., Leter, E.M., Douwes, J.M., Van Dijk, A., Vonk-Noordegraaf, A., Dijk-Bos, K.K., Hoefsloot, L.H., Hoendermis, E.S., et al. (2013). TBX4 mutations (small patella syndrome) are associated with childhood-onset pulmonary arterial hypertension. *J. Med. Genet.* 50, 500–506. <https://doi.org/10.1136/JMEDGENET-2012-101152>.
7. Stankiewicz, P., Sen, P., Bhatt, S.S., Storer, M., Xia, Z., Bejjani, B.A., Ou, Z., Wisniewska, J., Driscoll, D.J., Maisenbacher, M.K., et al. (2009). Genomic and Genic Deletions of the FOX Gene Cluster on 16q24.1 and Inactivating Mutations of FOXF1 Cause Alveolar Capillary Dysplasia and Other Malformations. *Am. J. Hum. Genet.* 84, 780–791. <https://doi.org/10.1016/j.ajhg.2009.05.005>.
8. Whitsett, J.A. (2010). Review: The intersection of surfactant homeostasis and innate host defense of the lung: lessons from newborn infants. *Innate Immun.* 16, 138–142. <https://doi.org/10.1177/1753425910366879>.
9. Cheng, Q., Wei, T., Farbiak, L., Johnson, L.T., Dilliard, S.A., and Siegwart, D.J. (2020). Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR–Cas gene editing. *Nat. Nanotechnol.* 15, 313–320. <https://doi.org/10.1038/s41565-020-0669-6>.
10. Wang, X., Liu, S., Sun, Y., Yu, X., Lee, S.M., Cheng, Q., Wei, T., Gong, J., Robinson, J., Zhang, D., et al. (2023). Preparation of selective organ-targeting (SORT) lipid nanoparticles (LNPs) using multiple technical methods for tissue-specific mRNA delivery. *Nat. Protoc.* 18, 265–291. <https://doi.org/10.1038/s41596-022-00755-x>.
11. Sun, Y., Chatterjee, S., Lian, X., Traylor, Z., Sattiraju, S.R., Xiao, Y., Dilliard, S.A., Sung, Y.C., Kim, M., Lee, S.M., et al. (2024). In vivo editing of lung stem cells for durable gene correction in mice. *Science* 384, 1196–1202. <https://doi.org/10.1126/SCIENCE.ADK9428>.
12. Zeng, G., He, Z., Yang, H., Gao, Z., Ge, X., Liu, L., Liu, Z., and Chen, Y. (2024). Cationic Lipid Pairs Enhance Liver-to-Lung Tropism of Lipid Nanoparticles for In Vivo mRNA Delivery. *ACS Appl. Mater. Inter.* 16, 25698–25709. <https://doi.org/10.1021/ACSAMI.4C02415>.
13. Eygeris, Y., Henderson, M.I., Curtis, A.G., Jozić, A., Stoddard, J., Reynaga, R., Chirco, K.R., Su, G.L.N., Neuringer, M., Lauer, A.K., et al. (2024). Preformed Vesicle Approach to LNP Manufacturing Enhances Retinal mRNA Delivery. *Small* 20, e2400815. <https://doi.org/10.1002/SMLL.202400815>.
14. Kaczmarek, J.C., Patel, A.K., Kauffman, K.J., Fenton, O.S., Webber, M.J., Heartlein, M.W., DeRosa, F., and Anderson, D.G. (2016). Polymer-Lipid Nanoparticles for Systemic Delivery of mRNA to the Lungs. *Angew. Chem. Int. Ed. Engl.* 55, 13808–13812. <https://doi.org/10.1002/ANIE.201608450>.
15. Kaczmarek, J.C., Kauffman, K.J., Fenton, O.S., Sadler, K., Patel, A.K., Heartlein, M.W., DeRosa, F., and Anderson, D.G. (2018). Optimization of a Degradable Polymer-Lipid Nanoparticle for Potent Systemic Delivery of mRNA to the Lung Endothelium and Immune Cells. *Nano Lett.* 18, 6449–6454. <https://doi.org/10.1021/acs.nanolett.8b02917>.
16. Kaczmarek, J.C., Patel, A.K., Rhym, L.H., Palmiero, U.C., Bhat, B., Heartlein, M.W., DeRosa, F., and Anderson, D.G. (2021). Systemic delivery of mRNA and DNA to the lung using polymer-lipid nanoparticles. *Biomaterials* 275, 120966. <https://doi.org/10.1016/j.biomaterials.2021.120966>.
17. Mastorakos, P., da Silva, A.L., Chisholm, J., Song, E., Choi, W.K., Boyle, M.P., Morales, M.M., Hanes, J., and Suk, J.S. (2015). Highly compacted biodegradable DNA nanoparticles capable of overcoming the mucus barrier for inhaled lung gene therapy. *Proc. Natl. Acad. Sci. USA* 112, 8720–8725. <https://doi.org/10.1073/pnas.1502281112>.
18. Capasso Palmiero, U., Kaczmarek, J.C., Fenton, O.S., and Anderson, D.G. (2018). Poly( $\beta$ -amino ester)-co-poly(caprolactone) Terpolymers as Nonviral Vectors for mRNA Delivery In Vitro and In Vivo. *Adv. Healthc. Mater.* 7, 1800249. <https://doi.org/10.1002/ADHM.201800249>.
19. Jain, S., Kumar, S., Agrawal, A.K., Thanki, K., and Banerjee, U.C. (2013). Enhanced transfection efficiency and reduced cytotoxicity of novel lipid-polymer hybrid nanoparticles. *Mol. Pharm.* 10, 2416–2425. <https://doi.org/10.1021/MP400036W>.
20. Karlsson, J., Rhodes, K.R., Green, J.J., and Tzeng, S.Y. (2020). Poly( $\beta$ -amino ester)s as gene delivery vehicles: challenges and opportunities. *Expert Opin. Drug Deliv.* 17, 1395–1410. <https://doi.org/10.1080/17425247.2020.1796628>.
21. Wang, C., Zhang, Y., and Dong, Y. (2021). Lipid Nanoparticle-mRNA Formulations for Therapeutic Applications. *Acc. Chem. Res.* 54, 4283–4293. <https://doi.org/10.1021/ACS.ACCOUNTS.1C00550>.
22. Madisen, L., Zwingman, T.A., Sunkin, S.M., Oh, S.W., Zariwala, H.A., Gu, H., Ng, L.L., Palmer, R.D., Hawrylycz, M.J., Jones, A.R., et al. (2010). A robust and high-throughput Cre reporting and characterization system for the whole mouse brain. *Nat. Neurosci.* 13, 133–140. <https://doi.org/10.1038/NN.2467>.
23. Kim, J., Jozic, A., Lin, Y., Eygeris, Y., Bloom, E., Tan, X., Acosta, C., MacDonald, K.D., Welscher, K.D., and Sahay, G. (2022). Engineering Lipid Nanoparticles for Enhanced Intracellular Delivery of mRNA through Inhalation. *ACS Nano* 16, 14792–14806. <https://doi.org/10.1021/ACS.NANO.2C05647>.
24. van der Meel, R., Grisoni, F., and Mulder, W.J.M. (2024). Lipid discovery for mRNA delivery guided by machine learning. *Nat. Mater.* 23, 880–881. <https://doi.org/10.1038/s41563-024-01934-9>.
25. Karikó, K., Buckstein, M., Ni, H., and Weissman, D. (2005). Suppression of RNA recognition by Toll-like receptors: The impact of nucleoside modification and the evolutionary origin of RNA. *Immunity* 23, 165–175. <https://doi.org/10.1016/j.immuni.2005.06.008>.
26. Nance, K.D., and Meier, J.L. (2021). Modifications in an Emergency: The Role of N1-Methylpseudouridine in COVID-19 Vaccines. *ACS Cent. Sci.* 7, 748–756. <https://doi.org/10.1021/ACSCENTSCI.1C00197>.
27. Izunobi, J.U., and Higginbotham, C.L. (2011). Polymer molecular weight analysis by <sup>1</sup>H NMR spectroscopy. *J. Chem. Educ.* 88, 1098–1104. <https://doi.org/10.1021/ED100461V>.
28. Rindler, T.N., Stockman, C.A., Filuta, A.L., Brown, K.M., Snowball, J.M., Zhou, W., Veldhuizen, R., Zink, E.M., Dautel, S.E., Clair, G., et al. (2017). Alveolar injury and regeneration following deletion of ABCA3. *JCI Insight* 2, e97381. <https://doi.org/10.1172/JCI.INSIGHT.97381>.
29. DiGiovanni, G.T., Han, W., Sherrill, T.P., Taylor, C.J., Nichols, D.S., Geis, N.M., Singha, U.K., Calvi, C.L., McCall, A.S., Dixon, M.M., et al. (2023). Epithelial Yap/Taz are required for functional alveolar regeneration following acute lung injury. *JCI Insight* 8, e173374. <https://doi.org/10.1172/JCI.INSIGHT.173374>.
30. Vila Ellis, L., Cain, M.P., Hutchison, V., Flodby, P., Crandall, E.D., Borok, Z., Zhou, B., Ostrin, E.J., Wythe, J.D., and Chen, J. (2020). Epithelial Vegfa Specifies a Distinct Endothelial Population in the Mouse Lung. *Dev. Cell* 52, 617–630.e6. <https://doi.org/10.1016/j.devcel.2020.01.009>.
31. Zhou, L., Lim, L., Costa, R.H., and Whitsett, J.A. (1996). Thyroid transcription factor-1, hepatocyte nuclear factor-3beta, surfactant protein B, C, and Clara cell secretory protein in developing mouse lung. *J. Histochem. Cytochem.* 44, 1183–1193. <https://doi.org/10.1177/44.10.8813084>.
32. Little, D.R., Lynch, A.M., Yan, Y., Akiyama, H., Kimura, S., and Chen, J. (2021). Differential chromatin binding of the lung lineage transcription factor NKX2-1 resolves opposing murine alveolar cell fates in vivo. *Nat. Commun.* 12, 2509. <https://doi.org/10.1038/s41467-021-22817-6>.
33. Stripp, B.R., Lund, J., Mango, G.W., Doyen, K.C., Johnston, C., Hultenby, K., Nord, M., and Whitsett, J.A. (1996). Clara cell secretory protein: a determinant of PCB bioaccumulation in mammals. *Am. J. Physiol.* 271, L656–L664. <https://doi.org/10.1152/AJPLUNG.1996.271.4.L656>.
34. Tan, Y., Huang, Z., Jin, Y., Wang, J., Fan, H., Liu, Y., Zhang, L., Wu, Y., Liu, P., Li, T., et al. (2024). Lipid droplets sequester palmitic acid to disrupt endothelial ciliation and exacerbate atherosclerosis in male mice. *Nat. Commun.* 15, 8273. <https://doi.org/10.1038/s41467-024-52621-x>.
35. Narvaez Del Pilar, O., Gacha Garay, M.J., and Chen, J. (2022). Three-axis classification of mouse lung mesenchymal cells reveals two populations of myofibroblasts. *Development* 149, dev200081. <https://doi.org/10.1242/DEV.200081>.